

Monotonicity of Quantum Entropy

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Main Result

Theorem (Beigi & Mehrabi, 2025)

Let ρ be an m -mode bosonic quantum state. Then

$$S(\rho^{\boxplus(n+1)}) \geq S(\rho^{\boxplus n})$$

Hilbert Spaces

Hilbert Space

A complex Hilbert space, $\mathcal{H}$ is vector space over $\mathbb{C}$ equipped with an inner product

$$\langle \cdot | \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$$

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We typically write elements of $\mathcal{H}$ as $|\phi\rangle$. And, every element of $\mathcal{H}$ gives a linear functional $\langle \phi| : \mathcal{H} \rightarrow \mathbb{C}$ by $|\psi\rangle \mapsto \langle \phi|\psi\rangle$.

Operators

If f is a function, let $D(f)$ be the domain of f .

Bounded and Unbounded Operators

A linear operator $A : D(A) \subset \mathcal{H} \rightarrow \mathcal{H}$ is called bounded if $D(A) = \mathcal{H}$ and

$$\sup_{\|x\|=1} \|Ax\| < \infty$$

and is called unbounded otherwise.

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Unbounded operators appear in Quantum Mechanics all the time. Two which pop up very frequently are called the position and momentum operators on $L^2(\mathbb{R}^2)$ given by

$$q = x$$

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$$q = x$$

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These operators are unbounded. Not only do they not have full domain, they also do not satisfy the supremum of norm on unit ball being finite.

Operators

Adjoint

If A is a bounded operator on a Hilbert space, then there exists a unique bounded operator A^* called the adjoint of A such that

$$\langle x|Ay\rangle = \langle A^*x|y\rangle$$

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These definitions can be extended to unbounded operators with care.

Trace

Let A be a positive bounded operator on $\mathcal{H}$ and $\{e_i\}_{i=1}^{\infty}$ be an o.n.b. of $\mathcal{H}$.

Trace

The trace of A is given by

$$\mathrm{Tr}(A) = \sum_{i=1}^{\infty} \langle e_i | A e_i \rangle$$

Moreover, if A is an operator such that $\mathrm{Tr}(|A|) = \mathrm{Tr}(\sqrt{AA^*}) < \infty$, then we call A trace class.

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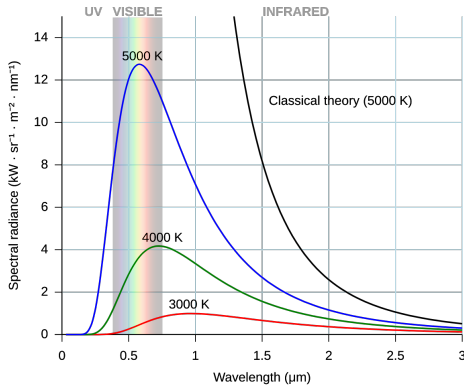
Note: any positive operator with finite trace is automatically trace class.

Quantum Mechanics

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Quantum Mechanics

Schrödinger Equation

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) \right] \Psi(x, t)$$

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Quantum States

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In “Classical” QM, states are given by wave functions which are norm 1, $L^2(\mathbb{R}^2)$ functions that satisfy Schrödinger’s equation. So, if $\Psi(x, t) \in L^2(\mathbb{R}^2)$, then, for a fixed time t , the Born rule tells us that

$$\int_a^b |\Psi(x, t)|^2 dx$$

gives us the probability of finding a particle in the interval $[a, b]$.

Operator QM

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Density Operator

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Density Operator

Let $\mathcal{H}$ be a Hilbert space. A positive, self-adjoint operator ρ on $\mathcal{H}$ of trace 1 is called a density operator.

Now, how do we "measure" our state? How can we determine the probability of some quantum object falling inside some bounds with this picture?

Spectral Projections

The idea is the following: we want an operator to represent some observable information about a quantum state (e.g. its momentum or position). And then, we use some operator determined by the observable operator and some Borel subset of the real line to make a determination of the probability of our state falling into this Borel set.

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Borel Functional Calculus

Let A be an operator on a Hilbert Space, and f be any Borel measurable function. Then, we can define the operator $f(A)$.

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Spectral Projection

For an operator A and Borel set B , then define $E_A(B) = \chi_B(A)$. Then $E_A(B)$ is an orthogonal projection on $\mathcal{H}$. Moreover, $E_A(\emptyset) = 0$ and $E_A(\mathbb{R}) = 1$.

Quantum Statistics

Let A be an operator on a Hilbert Space, ρ be a state, and B be a Borel set. Then, the probability of measuring ρ in B with respect to A is given by

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So now, we have constructed probability spaces from quantum mechanical states.

CCR Relations

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Uniqueness Theorem

Given a Hilbert space $\mathcal{H}$ and two operators P and Q satisfying the CCR relationship, then there exists a unique unitary mapping

$$T : \mathcal{H} \rightarrow L^2(\mathbb{R}) \otimes \mathcal{H}'$$

such that

$$\exp(ixP) = T^{-1}(\exp(ixp) \otimes 1_{\mathcal{H}'})T$$

and

$$\exp(iyQ) = T^{-1}(\exp(iyq) \otimes 1_{\mathcal{H}'})T$$

for $x, y \in \mathbb{R}$.

Central Limit Theorem

Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space. A (real) random variable is a measurable function $f : \Omega \rightarrow \mathbb{R}$. The probability a random variable has an outcome in an open set U is given by $\mathbb{P}(X^{\leftarrow}(U))$. And, the cumulative probability distribution $F_X(x)$ is given by

$$F_X(x) = \mathbb{P}(X^{\leftarrow}((-\infty, x)))$$

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Central Limit Theorem

Let $\{X_i\}_{i=1}^{\infty}$ be a collection of independent, identically distributed random variables. Then, the sequence

$$\hat{X}_n = \frac{\sum_{i=1}^n X_i}{\sqrt{n}}$$

converges in distribution to a Gaussian.

Monotonicity of Entropy

Entropy

Let X be a real valued random variable with probability density $f(x)$. Define the (Shannon) entropy of X to be

$$H(X) = - \int_{\mathbb{R}} f(x) \log(f(x)) dx = \mathbb{E}[-\log(X)]$$

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Monotonicity of Entropy (Artstein, Ball, Barthe, and Naor, 2004)

Let $\{X_i\}_{i=1}^{\infty}$ be a sequence of i.i.d. r.v.'s. Then

$$H(\hat{X}_{n+1}) \geq H(\hat{X}_n)$$

Quantum CLT

Quantum Central Limit Theorem (Cushen and Hudson, 1971)

Let $\{\rho_i\}$ be a sequence of independent and identically distributed states, and let $\hat{\rho}_n = \frac{\sum_{i=1}^n \rho_i}{\sqrt{n}}$. Then

$$\hat{\rho}_n \rightarrow \rho_\Gamma$$

where ρ_Γ is a Gaussian state.

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QCLT Definitions

(Weak) Convergence of States

Let $\{\rho_n\}_{n=1}^{\infty}$ and ρ be states on $\mathcal{H}$. Then $\rho_n \rightarrow \rho$ if and only if for every operator A on $\mathcal{H}$,

$$\mathrm{Tr}(A\rho_n) \rightarrow \mathrm{Tr}(A\rho)$$

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Intuitively, what does this say w.r.t. our previous slides? Next we want to talk about the sum of states.

Quasi-Characteristic Function

Let $x, y \in \mathbb{R}$, and ρ a state on $L^2(\mathbb{R})$. Then the quasi-characteristic function of ρ is given by

$$f_{\rho}(x, y) = \text{Tr}(\exp(i(xp + yq))\rho)$$

Gaussian States

What is the analog of a Gaussian distribution to QM?

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Gaussian States

A state ρ is called Gaussian if $f_\rho(x, y)$ looks like a Gaussian distribution.

Of course, this is entirely too general, but hopefully it just gives you taste that unlike in classical statistics where there is one single defined Gaussian (of mean 0, variance 1), the world of QM is full of things that look “Gaussian.”

QCF and States

The characteristic function is actually simply called the Wigner transform of a state. The inverse being the Weyl transform.

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Weyl-Wigner Correspondence

There exists a unitary mapping Δ from $L^2(\mathbb{R}^2)$ to Hilbert-Schmidt operators on $L^2(\mathbb{R})$ given by

$$\Delta(F) = (2\pi)^{-2} \int_{\mathbb{R}^2} F(a, b) \exp(i(ap + bq)) dadb$$

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This theorem tells us that every state is uniquely determined by its Quasi-Characteristic function.

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to be the unique state determined by

$$\Delta^{-1} \left(f_{\rho_1} \left(\frac{v}{\sqrt{2}} \right) f_{\rho_2} \left(\frac{v}{\sqrt{2}} \right) \right)$$

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for $v \in \mathbb{R}^2$. Then, we can easily extend this to arbitrary n . Hence, we define the n -fold convolution to be

$$\rho^{\boxplus n} = \frac{\sum_{i=1}^n \rho}{\sqrt{n}}$$

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von Neumann Entropy

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Beigi and Mehrabi, 2025

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Why is this such a big deal? It is the culmination of almost 100 years of the combination of classical statistics, classical information theory, quantum theory development, and quantum information theory.

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4. Stone-von Neumann theorem and evolutions of quantum states via unitary operators
5. (Symmetric) Fock Space representation, $\mathcal{F}_+ = \bigoplus_{n=0}^{\infty} \mathcal{H}_+^n$
6. Theory of unbounded operators

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